

A GUIDE FOR UAE VILLA HOMEOWNERS

# Why Water Pressure Drops.

## Ten common problems, and how to solve them.

Weak water pressure in a UAE villa almost always comes down to one of ten things. This guide starts with how to notice a pressure problem at all, explains each cause in plain language, tells you what to check yourself, and shows you honestly where a Grundfos SCALA2 booster is the answer, and where the fix is a clean, a repair or a valve.

**Where some see a problem, we see a possibility to make things better.**

How to notice a problem

Meet the ten problems

Find your problem fast

How SCALA2 helps

A 2-minute check

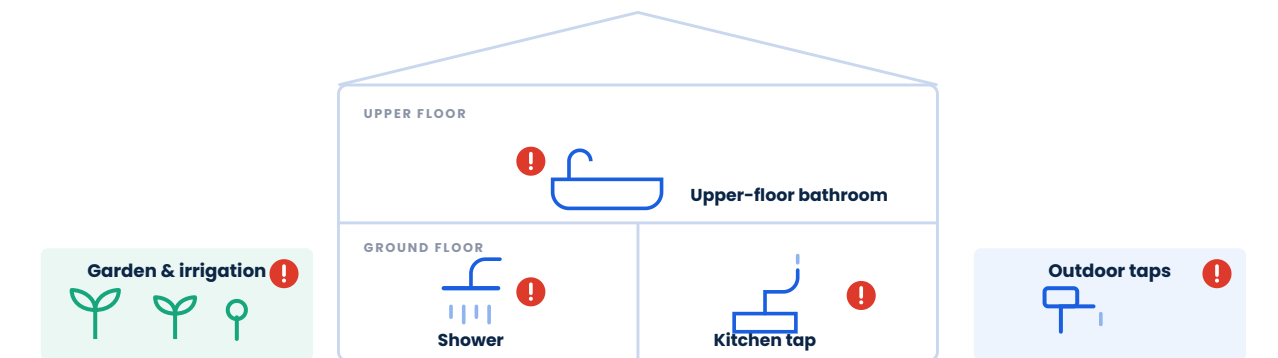


# You may have a pressure issue if any of this sounds familiar.

Most homeowners live with weak pressure for months before naming it, because it arrives gradually and every tap has its own excuse. These are the six signs worth taking seriously. One of them is enough to start looking.

- 1 Your shower feels weaker than usual
- 2 Taps take longer to fill sinks or containers
- 3 Upper-floor bathrooms receive less water
- 4 Outdoor taps and irrigation systems have weaker water flow
- 5 Water pressure drops when another tap is opened
- 6 Pressure changes throughout the day

## WHERE IT SHOWS UP IN A VILLA



## START WITH THE PATTERN, NOT THE TAP

Where and when it happens matters more than how bad it feels. One tap or the whole villa? Better downstairs than up? Only when the house is busy? Gradual over years, or sudden after a service visit? Those four answers point straight at the cause, and the next pages take each one in turn.



# Weak pressure is not bad luck. It is a fixable problem.

**And usually, the way it behaves tells you exactly what is wrong.**

If your shower fades when someone runs the kitchen tap, or the upstairs bathroom never feels as strong as the one downstairs, you are not imagining it. In most villas the cause is built into the plumbing, or has quietly accumulated inside it. The good news is that each cause has a clear set of signs, and once you know which one you are facing, the fix is usually straightforward.

## The ten problems, at a glance

### A The system cannot keep up



#### PROBLEM 1

Pressure drops when demand increases



#### PROBLEM 2

Pipes that are too small



#### PROBLEM 3

One pipe for the whole villa

### B Something is restricting the flow



#### PROBLEM 4

Blocked at the outlet



#### PROBLEM 5

Clogged filters and strainers



#### PROBLEM 6

Pipe scale build-up



#### PROBLEM 7

Partially closed valves

### C Something is wrong in the system



#### PROBLEM 8

Hidden water leak



#### PROBLEM 9

Only the hot water is weak



#### PROBLEM 10

Air trapped in the pipes

#### HOW TO USE THIS GUIDE

Read the ten problems in their three families, then turn to the diagnosis page to match what you are seeing to the likely cause. Every problem tells you what you notice, what could be causing it, what you can check yourself, and the recommended solution, and ends with an honest note on how SCALA2 helps, or where the answer is a clean, a repair or a valve instead.

#### A QUICK WORD ON SCALA2

SCALA2 is the Grundfos all-in-one home booster pump. It sits after your roof tank and gives every tap one steady pressure, adapting continuously to what your home is asking for. We point to it only where it genuinely solves the problem, and you will meet it properly at the end.



# When the house gets busy, the pressure gives way.

These are the problems of capacity: the supply is fine until the household asks it for more than it was built to give. This is where a booster earns its place.



## PROBLEM 1

### Water pressure drops when demand increases

The most common villa issue of all, and the one people most often mistake for bad luck.

#### WHAT YOU NOTICE

- Pressure drops when multiple taps are used
- Showers weaken when washing machines or irrigation systems are operating

#### WHAT COULD BE CAUSING IT

Your family's water demand may exceed the capability of the existing pressure system. Common examples: multiple bathrooms, garden irrigation, swimming pool filling, laundry equipment, and several taps operating simultaneously.

#### WHAT YOU CAN CHECK

- How many taps, showers, or appliances are being used at the same time
- Any recent home renovations or villa extensions
- New appliances or systems that use water regularly

#### RECOMMENDED SOLUTION

Upgrade to a pressure boosting solution that automatically adapts to variable demand.

#### OUTLETS OPEN AT THE SAME TIME

1 tap



**Strong**

Kitchen tap only

+ 2 showers



**Fading**

Two bathrooms in use

+ laundry

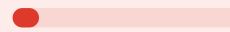


**Weak**

Washing machine on

#### PRESSURE AT YOUR SHOWER

+ irrigation



**Trickle**

Demand > capacity

#### Yes, SCALA2 solves this

This is exactly what SCALA2 is made for. The moment someone opens another tap, it instantly tops the pressure back up, so your shower stays strong even when the kitchen and laundry are running too. Up to 8 taps at once, with no fade.



## PROBLEM 2

### Pipes that are too small

Narrow or long, undersized pipes, common in older or cost-built villas, cannot carry enough water once more than one tap is open.

#### WHAT YOU NOTICE

- Good pressure at one tap, then a clear drop the instant a second or third opens
- Strong on its own, weak the moment the house gets busy

#### WHAT COULD BE CAUSING IT

The pipework itself is the bottleneck. It was sized for a quieter house than the one you live in, so the pressure sags as soon as demand rises.

#### WHAT YOU CAN CHECK

- Whether one tap alone is strong but several together are weak
- The age and build standard of the villa
- Whether long pipe runs reach the far side of the plot

#### RECOMMENDED SOLUTION

A booster that raises and holds pressure across the whole system, rather than replacing every pipe.

#### Yes, SCALA2 solves this

SCALA2 lifts and holds the pressure the pipework can no longer deliver on its own, so your shower stays strong even when the kitchen and laundry are running too.



# When the pressure fades floor by floor.



## PROBLEM 3

### One pipe for the whole villa

Many villas have one set of pipes feeding the whole house, with nothing to give the upper floors their own boost.

#### WHAT YOU NOTICE

- Pressure that is clearly weaker the higher up the villa you go
- The upstairs shower never feels as strong as the one downstairs

#### WHAT COULD BE CAUSING IT

Every tap shares the same pressure, so the higher up you go, the weaker it gets. Height is working against you and nothing in the system is compensating for it.

#### WHAT YOU CAN CHECK

- The same fixture on each floor, one after the other
- How many floors the supply has to lift water to
- The top floor at your household peak, not at a quiet hour

#### RECOMMENDED SOLUTION

**A booster sized for the height of the home, so every floor is served at the same pressure.**

**Yes, SCALA2 solves this**

A perfect fit. SCALA2 lifts the whole villa to one strong, even pressure and holds it there, so the upstairs shower feels just as good as the one downstairs. It is built for homes up to 3 floors and 8 taps, which covers most villas.



## PROBLEM 4

### Blocked at the outlet

When only one shower or tap is weak and the rest of the home is fine, the blockage is right there at that outlet.

#### WHAT YOU NOTICE

- Weak or patchy flow at one shower only
- A slow single tap or appliance, while everything else is normal

#### WHAT COULD BE CAUSING IT

Showerheads fur up with limescale, and the small filters on kitchen and utility taps slowly clog.

#### WHAT YOU CAN CHECK

- Whether it affects one outlet only, with the rest of the house running as usual
- The showerhead or tap aerator, unscrewed and held up to the light
- Whether flow returns when the head is removed

#### RECOMMENDED SOLUTION

**Clean the showerhead or swap the blocked filter and the flow comes back.**

**A quick fix, not a pump job**

SCALA2 keeps the rest of your home at full, steady pressure, so it is easy to see the trouble is only that one spot.



# When the flow is being squeezed on the way.



## PROBLEM 5

### Clogged filters and strainers

The slow one. Nothing breaks, nothing changes overnight, and the house quietly gets weaker for months.

#### WHAT YOU NOTICE

- Water pressure has gradually reduced over time
- Water flow seems weaker than before
- Multiple taps, showers, or appliances may be affected
- The change happens gradually rather than suddenly

#### WHAT COULD BE CAUSING IT

Over time, sand, sediment, rust particles, or other debris can accumulate in filters and strainers. This restricts water flow and reduces the amount of water reaching your fixtures.

#### WHAT YOU CAN CHECK

- Inspect filters for dirt or sediment build-up
- Check when the filters were last cleaned or replaced
- Review the maintenance schedule for your water system

#### RECOMMENDED SOLUTION

Clean or replace filters according to the manufacturer's recommendations. Regular filter maintenance helps maintain water flow and supports optimal pump performance.

#### CLEAN FILTER



#### BLOCKED FILTER



#### Maintenance, not a pump job

No pump can clean a filter for you. But because SCALA2 normally holds your pressure rock steady, a slow decline is far easier to notice, and it protects the pump from running against a restriction it should not have to fight.



## PROBLEM 6

### Pipe scale build-up

The very slow one. This is hard water doing its work quietly, over years rather than months.

#### WHAT YOU NOTICE

- Ongoing pressure reduction over several years
- Certain bathrooms or taps affected more than others

#### WHAT COULD BE CAUSING IT

Mineral deposits can accumulate inside pipework, restricting water flow. The pipe is still there, but the space inside it is not.

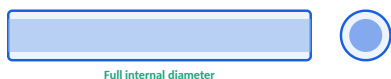
#### WHAT YOU CAN CHECK

- Age of plumbing system
- Areas with recurring flow issues
- Signs of hard water scaling

#### RECOMMENDED SOLUTION

Professional pipe cleaning, replacement, or plumbing assessment.

#### NEW PIPE



#### SCALED PIPE



#### A plumbing job, not a pump job

Scale has to be removed, and no booster can widen a pipe. SCALA2 gives the remaining pipework the strongest, steadiest feed it can have, and once the pipes are cleared, your home has both the space and the pressure it needs.



# When it changed suddenly, or over years.



## PROBLEM 7

### Partially closed valves

The cheapest problem in this guide to fix, and the easiest one to overlook entirely.

#### WHAT YOU NOTICE

- Sudden pressure reduction after maintenance work
- One area of the villa affected more than others

#### WHAT COULD BE CAUSING IT

A valve may not have been fully reopened after servicing. Nothing is broken. Something is simply not all the way back on.

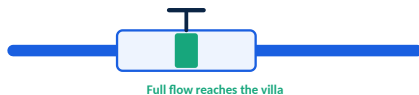
#### WHAT YOU CAN CHECK

- The main water shut-off valve to your home
- Valves connected to your water pressure boosting system
- Any water control valves serving specific areas of the villa

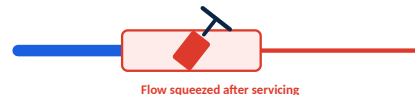
#### RECOMMENDED SOLUTION

Ensure all system valves are fully open and operating correctly.

FULLY OPEN



PARTIALLY CLOSED



#### Check this before anything else

Worth ruling out before you call anyone. SCALA2 will hold pressure up to whatever the valve allows through, so a half-closed valve is best found and opened first, and then the pump can do its job properly.



## PROBLEM 8

### Hidden water leak

A leak bleeds pressure out of the system before the water ever reaches your taps, and it quietly pushes up your water bill.

#### WHAT YOU NOTICE

- Increased water bills
- Pump running when no water is being used
- Damp patches on walls or floors
- Weak pressure across the whole villa, not one tap

#### WHAT COULD BE CAUSING IT

Water may be escaping from hidden pipework underground or inside walls.

#### WHAT YOU CAN CHECK

- Check for visible dampness
- Monitor water meter activity, it should not move with every tap closed
- Listen for running water sounds

#### RECOMMENDED SOLUTION

Arrange a professional leak detection inspection as early as possible.



#### Needs a repair, and early warning

A leak needs a plumber to find and fix, no pump can do that. But SCALA2 helps you catch it sooner. Because it normally holds your pressure rock steady, a sudden drop across the whole home is an early warning that something is leaking.



# When it stutters, or only the hot is weak.

These are the problems of fault: something in the system is not behaving as it should, and no amount of pressure will settle it until it is put right.



## PROBLEM 9

### Only the hot water is weak

If the cold runs fine but the hot trickles, the problem is in the hot water system, not the whole villa.

#### WHAT YOU NOTICE

- Normal pressure on the cold, weak pressure on the hot, at the same tap

#### WHAT COULD BE CAUSING IT

Residue and scale build up inside the water heater and block the hot outlet.

#### WHAT YOU CAN CHECK

- Compare hot and cold at one tap, if only the hot is weak it is the heater
- When the heater was last serviced
- Whether every hot outlet is affected, or only the ones nearest the heater

#### RECOMMENDED SOLUTION

A heater service to clear the build-up inside it.

#### Service the heater

SCALA2 makes sure your heater is fed with strong, steady water in the first place, so once it is cleaned, your hot water performs at its best.



## PROBLEM 10

### Air trapped in the pipes

A pocket of air caught in the pipework interrupts the flow of water, so the pressure comes and goes.

#### WHAT YOU NOTICE

- Pressure that stutters, and sudden bursts of air and water spitting from the taps
- Spitting taps, especially just after work on the system

#### WHAT COULD BE CAUSING IT

It usually appears right after a tank refill or maintenance, or where the system is poorly sealed.

#### WHAT YOU CAN CHECK

- Whether it began right after a refill or a service visit
- Whether it settles after running the tap for a minute
- Whether one line is affected or the whole house

#### RECOMMENDED SOLUTION

Bleed the affected line. If it keeps returning, have the system checked for a poor seal.

#### SCALA2 helps

SCALA2 clears the air out by itself and gets the water flowing again after a refill, instead of getting stuck. Air trapped further along the pipes may still need a quick bleed, but the pump keeps itself going.





# Match what you see to the cause.

Start with what you notice. The pattern usually points straight to the cause, and the answer is often a page or two away rather than a call-out away.

| WHAT YOU ARE NOTICING                                   | THE LIKELY CAUSE                                        |
|---------------------------------------------------------|---------------------------------------------------------|
| ● Low pressure everywhere in the villa                  | The pump, the tank, or the main supply                  |
| ● Pressure is different from floor to floor             | The pipe layout, one supply feeds every floor           |
| ● Pressure drops when several taps run                  | Demand is above the system, the most common villa issue |
| ● Low pressure at one tap or shower only                | A blockage at that one tap                              |
| ● Only the hot water is weak                            | The water heater                                        |
| ● It fell away gradually over months or years           | Clogged filters, or scale inside the pipes              |
| ● It dropped suddenly after maintenance work            | A valve left partly closed, or trapped air              |
| ● Higher bill, damp patches, pump runs with no tap open | A hidden leak                                           |
| ● Pressure changes at certain times of day              | Peak demand, on the supply or in your own household     |

## So, can a SCALA2 solve it?

|                                         |       |                                                                             |
|-----------------------------------------|-------|-----------------------------------------------------------------------------|
| 1. Pressure drops when demand increases | Yes   | Adapts instantly, keeps every tap strong when several run.                  |
| 2. Pipes that are too small             | Yes   | Lifts and holds the pressure the pipework cannot deliver alone.             |
| 3. One pipe for the whole villa         | Yes   | Gives the whole home one strong, even pressure on every floor.              |
| 4. Blocked at the outlet                | No    | Just clean the head or swap the filter.                                     |
| 5. Clogged filters and strainers        | No    | Clean or replace them on schedule. SCALA2 makes the decline easier to spot. |
| 6. Pipe scale build-up                  | No    | Needs professional pipe cleaning or replacement.                            |
| 7. Partially closed valves              | No    | Reopen the valve fully. Free, and worth checking first.                     |
| 8. Hidden water leak                    | No    | Needs a repair. SCALA2 warns you with a sudden drop.                        |
| 9. Only the hot water is weak           | No    | Service the heater. SCALA2 feeds it strong, steady water.                   |
| 10. Air trapped in the pipes            | Helps | Clears trapped air at the pump by itself.                                   |

### A QUICK TIP BEFORE YOU CALL ANYONE

Note whether the problem is one tap or the whole villa, whether it changes by floor, whether it only happens when several taps run, and whether it came on gradually or overnight. Those four answers usually point to the cause, and they make any visit faster and cheaper.



# Intelligent pressure boosting for modern villas.

Unlike conventional pumps that operate at a fixed speed, SCALA2 continuously adapts to your home's water demand. It is the difference between a system that runs flat out for a half-open tap and one that simply gives your home the pressure the moment needs.

## WHEN DEMAND IS LOW

### One kitchen tap open

SCALA2 runs at a lower speed while holding comfortable pressure.

PUMP SPEED



PRESSURE

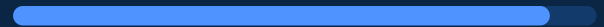


## WHEN DEMAND INCREASES

### Showers + kitchen + irrigation

SCALA2 automatically increases performance to keep pressure consistent.

PUMP SPEED



PRESSURE



## WHEN DEMAND IS LOW

### One kitchen tap open

SCALA2 operates at a lower speed while maintaining comfortable water pressure. Less noise, less power, same result at the tap.

## WHEN DEMAND INCREASES

### Multiple showers, kitchen tap, irrigation

SCALA2 automatically increases performance to keep pressure consistent throughout the villa, so the first shower does not pay for the second.

## THE RESULT

**Consistent shower experience**

**Reliable pressure on upper floors**

**Reduced pressure fluctuations**

**Quiet operation**

**Energy-efficient performance**



**In one compact unit**



# Meet SCALA2.

If the cause is a weak or shared supply, this is the part that fixes it.

## One quiet box does it all

SCALA2 sits after your roof tank and gives every tap and shower one steady pressure, even when several run at once. The pump, motor, tank, sensor, controller and valve all live inside a single compact unit, so there is nothing to piece together.



- Pump
- Motor
- Tank
- Sensor
- Controller
- Valve

Everything an old setup needed as separate parts, in one unit.

## What it does for your home

- **Steady pressure, always.** No more fading shower when the family opens other taps.
- **Great on every floor.** The upstairs shower finally feels as good as the one downstairs.
- **Lovely and quiet.** You will barely know it is running.
- **Strength you choose.** Set it once to the pressure your home likes.
- **Easier on bills.** It only uses the power the moment needs.
- **Simple and safe.** Fits in one go, and shuts itself off if anything is ever wrong.

## A 2-MINUTE CHECK, NO TOOLS NEEDED

Open one tap and notice how strong it is. Now open two or three more around the house at the same time and look back at the first. If the flow falls away noticeably, your supply cannot keep up when the home gets busy, and that is the everyday problem SCALA2 is built to solve.

## GOOD TO KNOW

SCALA2 works with the roof or ground tank you already have. If your home is bigger than 3 floors or 8 taps, Grundfos makes larger boosters for the same job.

Find the problem first. If it is your supply, SCALA2 gives every tap one steady, quiet, comfortable pressure. If it is a blockage, a filter, a valve, a leak or the heater, fix that, and SCALA2 keeps the rest of your home rock steady.

**Possibility in every drop . [grundfos.com/ae](https://grundfos.com/ae)**

**GRUNDFOS**   
Possibility in every drop